Lab#2: Data Definition and Addressing Modes.

Data Definition:

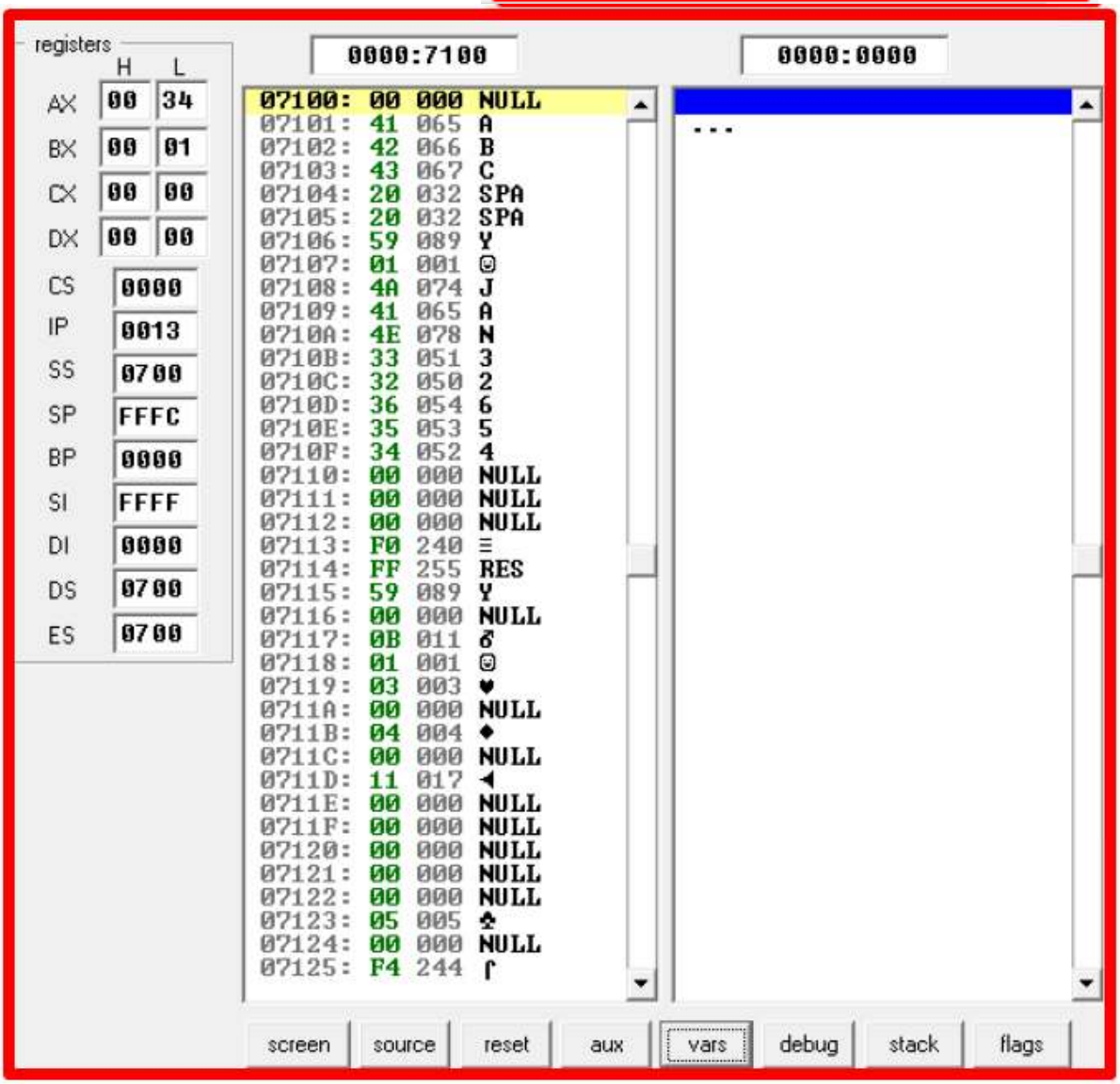
Assignment #1: Show the memory after defining the following variables:

```
ORG 100H
X1 DB ?
X2 DB 'ABC'
X3 DB 32
X4 DB 20H
X5 DB 01011001B
X6 DB 01, 'JAN'
X7 DB '32654'
X8 DB 3 DUP(0)

Y1 DW 0FFF0H
Y2 DW 01011001B
Y3 DW X7
Y4 DW 3, 4, 17
Y5 DW 2 DUP(0)
Y6 DW X8-X7
HLT
```

```
X1      00h
X2      41h
X3      20h
X4      20h
X5      59h
X6      01h
X7      33h
X8      00h
Y1      0FFF0h
Y2      0059h
Y3      010Bh
Y4      0003h
Y5      0000h
Y6      0005h
```

X1	00
X2	41
	42
	43
X3	20
X4	20
X5	59
X6	01
	4A
	41
	4E
X7	33
	32
	36
	35
	34
X8	00
	00
	00
Y1	F0
	FF
Y2	59
	00
Y3	0B
	01
Y4	03
	00
	04
	00
	11
	00
Y5	00
	00
	00
	00
Y6	05
	00



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Addressing Modes:

Assignment #2: Initializing the internal registers of the 80x86 as follows:

(AX) = 0000H
(BX) = 0001H
(CX) = 0002H
(DX) = 0003H
(SI) = 0010H
(DI) = 0020H
(BP) = 0030H
(DS) = 0B60H

```
01 ORG 100H
02
03 MOV AX, 0000H
04 MOV BX, 0001H
05 MOV CX, 0002H
06 MOV DX, 0003H
07 MOV SI, 0010H
08 MOV DI, 0020H
09 MOV BP, 0030H
10 MOV DS, 0B60H
11
12 HLT
```

Fill all memory locations in the range DS:00 through DS:1F with 00H and then initialize the following storage locations:

(DS:0001H) = BBBBH
(DS:0004H) = CCCCH
(DS:0011H) = DDDDH
(DS:0014H) = EEEEH
(DS:0016H) = FFFFH

```
01 ORG 100H
02
03 MOV AX, 0700H
04 MOV DX, AX
05 MOV SI, 0
06 L:MOV [SI], 00H
07 INC SI, 001FH
08 JBE L
09 MOV SI, 0001H
10 MOV [SI], 0BBBBH
11 MOV SI, 0004H
12 MOV [SI], 0CCCCH
13 MOV SI, 0011H
14 MOV [SI], 0DDDDH
15 MOV SI, 0014H
16 MOV [SI], 0EEEEH
17 MOV SI, 0016H
18 MOV [SI], 0FFFFH
19
20 HLT
```

Using emu8086 to assemble the instructions

(a) MOV AX, BX
(b) MOV AX, 0AAAAh
(c) MOV AX, [BX]
(d) MOV AX, [4]
(e) MOV AX, [BX+SI]
(f) MOV AX, [SI+4]
(g) MOV AX, [BX+SI+4]

Trace the execution of the instructions (a) through (g). Explain the execution of **each** instruction, including addressing mode, physical address for memory addressing mode, value in AX. Fill the table below.

Instruction	Addressing Mode	Physical Address	AX
a	Register A. M.	=	FFFF
b	Immediate A. M.	=	0B06
c	Indirect A. M.	0700:013D	AAAA
d	Direct A. M.	0700:013F	BBBB
e	Based Indexed A. M.	0700:0142	CCCC
f	Indexed A. M.	0700:0144	0000
g	Based Indexed A. M.	0700:0147	EEEE

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Solution:

```
01  ORG 100H
02  MOV AX, DS
03  MOV CX, 0020h
04  MOV DI, 0000h
05
06
07  MOV AL, 00H
08  REP STOSB
09
10
11  MOV WORD PTR [DI+0001h], 0BBBBh
12  MOV WORD PTR [DI+0004h], 0CCCCh
13  MOV WORD PTR [DI+0011h], 0DDDDh
14  MOV WORD PTR [DI+0014h], 0EEEEh
15  MOV WORD PTR [DI+0016h], 0FFFFh
16  HLT
```

Writing program

Given memory locations defined as below:

X DB 10,11,19,20,30,40,100,1,2,3,4,5,6

Write an assembler codes to replace values that are greater than 10 with zero values.

```
01  ORG 100H
02
03  X DB 10,11,19,20,30,40,100,1,2,3,4,5,6
04
05  MOV DI,0
06  MOV CX,13
07
08  L1: CMP X[DI],10
09  Jbe L2
10  zero: MOV X[DI],0
11  L2: INC DI
12  LOOP L1
13
14
15
16  HLT
```

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